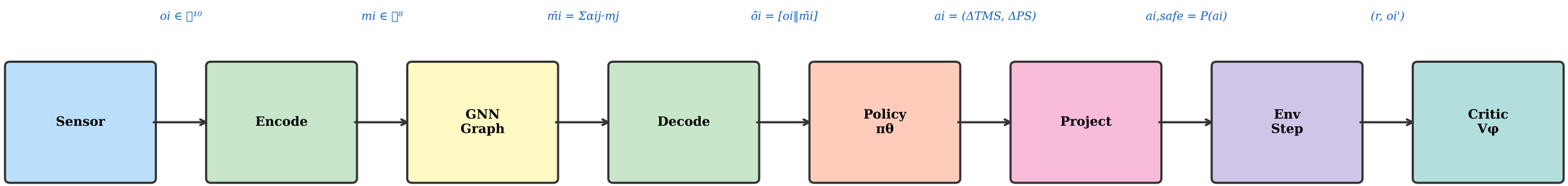
Figure 2: Data Flow Architecture (Single Training Step)



Local Observation (10-dim)

- V, I, P_dir, I_fault
- PV_level, topology_flag
- neighbor_TMS, neighbor_PS
- CTI_slack, time

GNN Attention

- αij = softmax(LeakyReLU(aᵀ[Whi‖Whj]))
- Message: 8 floats (32 bytes)
- Bandwidth: 0.256 kbps
- Graph: P/B pairs + same-bus

Safe RL Projection

- TMS ∈ [0.05, 1.1]
- PS ∈ [PS_min, PS_max]
- CTI lookahead binary search
- β* = max feasible scale

Centralized Critic

- s_global = [TMS_all‖PS_all‖
- PV_level‖topology‖fault_type]
- GAE(λ=0.95, γ=0.99)
- Value: Vφ(s_global)

Rollout Buffer: (oi, õi, ai,safe, r, oi', Vφ(s), log πθ(ai|õi))